Fernando Jr. Villanueva

juniorvillanueva156@outlook.com
https://fvjr156.github.io/my-portfolio

Passionate software developer with freelancing experience in crafting high-performance mobile applications using Java, Android Studio, and Room Databases, alongside scalable web applications powered by React, Express, MySQL, and Node.js.

Skills

- Backend Web Development
- Frontend Web Development
- Mobile App Development
- Research and Technical Documentation
- MySQL, MongoDB, SQL Server, SQLite, Room
- Git and GitHub
- Tailwind CSS and Bootstrap
- HTML and CSS Proficiency
- JavaScript and TypeScript Proficiency
- Java and C# Proficiency
- C/C++ and Low-Level Programming Familiarity
- MERN stack and MVC architecture familiarity

Work Experience

BridgeXSign: ML-Based Sign Language Translator Application
Contract Developer — June to December 2025

- Developed an Android application which uses a TFLite machine learning model to translate certain American Sign Language gestures into text and speech.
- Built a user-friendly and easy to navigate User Interface to allow users to quickly access different pages for translation, settings, and sign language gesture references.

ARISE: Augmented Reality PC Assembly and Disassembly Application
Contract Developer — May to August 2025

- Developed an Android application which uses Augmented Reality and 3D models to simulate PC assembly and disassembly processes.
- Implemented marker recognition for AR model spawning and rendering using Unity and Vuforia.

Education

Bachelor of Science in Computer Science
STI College Marikina
Expected finish March 2028

Years 1-2:

- Developed multi-platform applications using ASP.NET (web/mobile/desktop), Android Studio (Java), and Java AWT, targeting diverse devices and operating systems.
- Managed SQL Server databases using SQL Server and SQL Server Management Studio, with advanced queries, stored procedures, and database normalization.
- Implemented data structures (e.g. arrays, trees, graphs) and algorithms. Designed and analyzed efficient algorithms for real-world scenarios.
- Applied Object-Oriented Programming (OOP), Functional Programming, and MVC architecture in coding exercises and hands-

on projects, enhancing code modularity, scalability, and maintainability.

SHS — IT in Mobile App and Web Development Strand
STI College Marikina
August 2022 – July 2024

- Mastered 3 core programming languages (Java, C#, JavaScript) through hands-on projects and lessons, building foundational coding proficiency.
- Developed interactive web applications using the web technologies HTML, CSS, and JavaScript, implementing responsive web designs and smooth client-side functionality.
- Applied Object-Oriented Programming (OOP) and Functional programming concepts in hands-on projects, lessons, and coding exercises, optimizing code structure, modularity, scalability, and maintainability.
- Built one (1) mobile application using Android Studio and Java, and one (1) full-stack web-based application using MERN stack and MVC architecture, as parts of major course deliverables.
- Designed and queried relational databases using SQL Server and SQL Server Management Studio, executing CRUD operations and complex joins for data-driven projects.

Internship

Data Encoder Intern — Student-Trainee
National Bureau of Investigation, Marikina City Satellite Office
February to March 2024 (1 month)

- Requirement for SHS Work Immersion course
- Responsibilities:
  - Processed data for 500+ NBI clearance applicants (up to 50 applicants per day), ensuring 100% accuracy in record digitization.
  - Verified identities of 500+ applicants through submitted legal document cross-checks (e.g. birth certificates and past clearance applications), upholding data entry accuracy.
  - Guided 300+ applicants through clearance application processes, streamlining application submissions and resolving queries to improve turnaround times.
  - Contributed to a positive work environment by proactively assisting colleagues as needed, fostering strong professional relationships and a collaborative atmosphere.
  - Developed strong attention-to-detail skills through the constant review of entered information, ensuring high-quality results.
  - Used computer software to encode, store, and retrieve data.
  - Maintained strict and utmost confidentiality while handling sensitive client information according to company guidelines and legal regulations.

Certifications

- **Crash Course on Python**
  issued by Coursera — April 2026
  https://coursera.org/share/4d166512610af4021eee04ae9c640856

- **Foundations of Agile Project Management**
  issued by HackerRank — March 2026
  https://coursera.org/share/2592eec1c05cf07d832f044d8ff2de6f
- **React (Basic)**
  issued by HackerRank — November 2024
  https://www.hackerrank.com/certificates/6e931408ac0a
- **Java (Basic)**
  issued by HackerRank — November 2024
  https://www.hackerrank.com/certificates/19689f072134

Languages

- **English** – Fluent, Working Proficiency
- **Filipino (Tagalog)** - Native